Prediction of Adverse Drug Reactions - Using Advanced Statistical Methods

A Preprint

Arshad Ahamed Shajahan
School of Mathematics
University of Galway
Galway, Ireland
A.Shajahan2@universityofgalway.ie

Sai Shasank Dandu
School of Mathematics
University of Galway
Galway, Ireland
S.Dandu2@universityofgalway.ie

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Abstract

Adverse Drug Reactions represent a major challenge in pharmacovigilance and drug safety, contributing significantly to patient morbidity, mortality, and healthcare costs. Due to the high cost and time required to run clinical trials, there is significant interest in accurate computational methods that can aid in the prediction of ADRs for new drugs. So in this paper, a wider range of statistical models were developed and evaluated in predicting ADRs using two complementary sources of drug information, drug gene interaction profiles and chemical fingerprints. The models evaluated include a Naïve Baseline, Kernel Regression, Support Vector Machines with linear and RBF kernels, and a novel Weighted Non-negative Matrix Factorisation framework combined with Kernel Regression. Performance is assessed across 5-fold cross-validation using the Area Under the ROC Curve and Area Under the Precision-Recall Curve. Our results demonstrate that the Weighted NMF with Kernel Regression achieves the highest AUPR (0.5007 ± 0.0021), making it the most effective method for handling the imbalanced nature of ADR datasets, while VKR NMF achieves the highest AUROC (0.9183 ± 0.0025). An interactive web application was further developed enabling real time single drug ADR prediction.

1 Introduction

Adverse Drug Reactions (ADRs) are unintended, harmful responses to medications administered at normal therapeutic doses. ADRs are among the leading causes of hospitalisation and morbidity worldwide, with a pooled median prevalence of ADR-related hospital admissions estimated at 6.3% in developed countries and 5.5% in developing countries, based on a systematic review of studies published between 2000 and 2015 (Angamo et al. 2016).

Beyond the human cost, ADRs impose substantial financial burdens on healthcare systems. The same European Commission analysis estimated that 197,000 deaths per year in the EU are attributable to ADRs, with a total societal cost of €79 billion as of 2008 (Formosa et al. 2018). Traditional approaches to ADR detection rely primarily on post market pharmacovigilance, monitoring drug safety signals after a drug has already entered clinical use (Koyama et al. 2024). While these methods are effective, someone must experience an adverse reaction in real life to flag such behaviour. Hence, computational prediction of ADRs, offers the potential to anticipate safety issues before clinical exposure, informing drug development decisions and personalising treatment.

Drugs can be characterised by their interaction profiles with genes and by their chemical structure, both of which are expected to carry information about their biological activity and, by extension, their side effect profiles. Therefore, the drug gene profiles and chemical structure were used as primary inputs for the model in predicting ADR's. However, the construction of such models faces two fundamental statistical challenges:

1. **Sparsity**: The drug side effect matrix is extremely sparse. Most drug side effect pairs are unobserved, not because the side effect is absent, but because it has not been reported or studied.
2. **Class Imbalance**: The number of known positive drug side effect associations is vastly smaller than the number of negatives, making standard classifiers biased toward predicting the majority (negative) class.

Addressing these challenges requires careful model selection and evaluation. The Area Under the Receiver Operating Characteristic curve (AUROC) is a widely used metric that quantifies a classifier's ability to discriminate between positive and negative classes across all classification thresholds, calculated as the area under the curve plotting the true positive rate (sensitivity) against the false positive rate (1 - specificity). However, the ADR dataset possess a highly imbalanced classes where positive cases are rare relative to the negative cases.

The Area Under the Precision Recall Curve (AUPR) plots, precision (the proportion of predicted positives that are true positives) against recall (the proportion of actual positives correctly identified) across all thresholds. Since precision is sensitive to number of false positives, AUPR provides a more informative and discriminative assessment of the model performance. So, both AUPR and AUROC has been evaluated for all the models.

The previous research work from (Zhong et al. 2024) serves as a base for our research. It addressed the core issue with imbalanced datasets but introducing a novel method of using Non-Negative Matrix Factorization. In this research, the results from the previous paper is replicated and further extended the work by expanding the dataset and by introducing a novel method, Weighted Non-Negative Matrix Factorization for predicting ADRs using drug gene interaction (DGI) profiles and chemical fingerprints.

In the remainder of this paper, Section 2 describes the datasets, Section 3 … and so on.

2 Data

The primary data for this study was accessed through various publicly available sources. In particular, the SIDER 4.1 database was used to obtain drug side-effect information. SIDER ("Side Effect Resource") is a publicly available database that compiles information on marketed drugs and their recorded adverse drug reactions, extracted from package inserts and clinical trial data. the current release contains data on 1,430 drugs, 5,880 ADRs, and 140,064 drug–ADR pairs (Kuhn et al. 2016).

Additionally, the Drug–Gene Interaction Database (DGIdb 4.0) was used to obtain drug–gene interaction data. DGIdb is a web resource that aggregates information on drug–gene interactions and druggable genes from publications, databases, and other web-based sources, with drug, gene, and interaction data normalised and merged into conceptual groups (Freshour et al. 2021).

PubChem, a freely accessible chemical database maintained by the National Center for Biotechnology Information (NCBI), was used to retrieve drug chemical structure fingerprints. Mapping the drug names to side effects, gene interactions, and chemical fingerprints resulted in 973 unique data points across the three databases. Table 1 provides an overview of the dimensionality of the three databases.

All three datasets were binarised (binary matrices) prior to modelling. The DGI and Chemical fingerprints data had all the drugs in the first column. Genes and generated fingerprints were presented in each columns in the respective datasets. Table 1 summarises the dataset dimensions.

Table 1: Summary of dataset dimensions.

Feature Type	Dimensionality
Drug-Gene Interaction	$973 \times 3{,}647$
Chemical Fingerprints	$973 \times 1{,}024$
Side Effect Matrix	973×5779

2.1 Class Imbalance in the dataset

The drug side effect matrix is highly sparse, with a positive rate well below 10% for most side effects. This severe class imbalance motivates the use of AUPR as a primary evaluation metric alongside AUROC, since

AUROC can appear inflated in imbalanced settings. In the entire dataset, about 97% of the data represents the negative class and remaining as positive class

3 Methods

Initially exploratory data analysis, including descriptive statistics and heatmaps, was performed to ??????. Furthermore, to predict the ADR, six modelling approaches of increasing sophistication were employed. All models take as input either the DGI matrix, the chemical fingerprints matrix, or both, and output a probability score for each drug side effect pair. 5-fold cross validation was performed to assess the predictive ability of the proposed models. Due to high sparsity of the drug side effect matrix (with a positive rate well below 10% for most side effects), AUPR was used as a primary evaluation metric alongside AUROC, since AUROC can appear inflated in imbalanced settings. Below we briefly discuss the proposed modelling approaches.

1. **Naïve Baseline** The Naïve Baseline assigns to each drug-side effect pair the marginal prevalence of that side effect across the training set. This model ignores the drug gene interaction and chemical fingerprints data, and predicts probability based on the prevalence of side effects among the given dataset.
2. **Kernel Regression** Kernel Regression (KR) operates by mapping drugs into a high dimensional similarity space. For a new drug (d), its side effect probabilities are predicted as a weighted average of training drugs' side effect profiles, where weights are proportional to their similarity to new drug (d). This model projects the new drug into the same high-dimensional space as the training drugs and predicts its side effects based on its position in that space.
3. **Support Vector Machine (SVM)** The SVM models treat ADR prediction as a binary classification problem, training a separate classifier per side effect. We evaluated two kernel configurations: SVM-Linear: A linear SVM in the feature space, appropriate when the number of features exceeds the number of samples. SVM-RBF: An SVM with a Radial Basis Function (RBF) kernel, which maps the data into an infinite-dimensional feature space and finds a maximum-margin hyperplane there.
4. **Non-negative Matrix Factorisation with Kernel Regression (VKR NMF)** Non-negative Matrix Factorisation (NMF) decomposes the training side effect matrix into two lower-rank non-negative matrices:
$$Y \approx UH^\top, \quad U \in \mathbb{R}_+^{n\times k}, \quad H \in \mathbb{R}_+^{s\times k}$$
where min(n,s) is the number of latent features. The matrix 'U' captures the drug level latent representation (side effect profiles in latent space), and 'H' captures the side effect level loadings. For prediction, Kernel Regression is applied to project new drugs into the latent space:

$$\hat{U}_{d^*} = K(d^*, D_{\text{train}})W$$

where 'W' is a weight matrix learned by regressing 'U' onto the kernel matrix. The predicted side effects are then reconstructed as:

$$\hat{Y}_{d^*} = \hat{U}_{d^*}H^\top$$

This approach captures latent structure in the side effect matrix while leveraging drug similarity for prediction.

5. **Weighted NMF with Kernel Regression (VKR Weighted NMF)** Standard NMF treats all entries of 'Y' equally, which is problematic given sparsity and class imbalance. Weighted NMF addresses this by upweighting observed positive entries in the factorisation objective:

$$\min_{U\geq 0, H\geq 0} \sum_{i,j} W_{ij}\left(Y_{ij} - [UH^\top]_{ij}\right)^2$$

where $W_{ij} > 1$ for observed positive pairs and $W_{ij} = 1$ otherwise. This encourages the model to accurately reconstruct the rare positive associations that are most predictively valuable. Kernel Regression is then applied as in the standard NMF case to project new drugs. This model is designed specifically to address the imbalance problem, prioritising precision on positive predictions. The penalty factor of 1:10 is applied, for predicting all true outcomes wrongly. Since the dataset is imbalanced, this penalty adds weightage to predict all the true positives correctly.

4 Results

Performing Exploratory Data Analysis (EDA) in the dataset, brought out some key findings and understaandings about the ADR dataset. In Figure 1 you can see the drugs associated with the largest number of reported side effects. The results show that a relatively small number of ADRs account for a substantial proportion of all reported events, indicating a highly skewed distribution of side effects. The drugs mentioned below interacts with our central nervous system (CNS) and hence the large number of side effects associated with them.

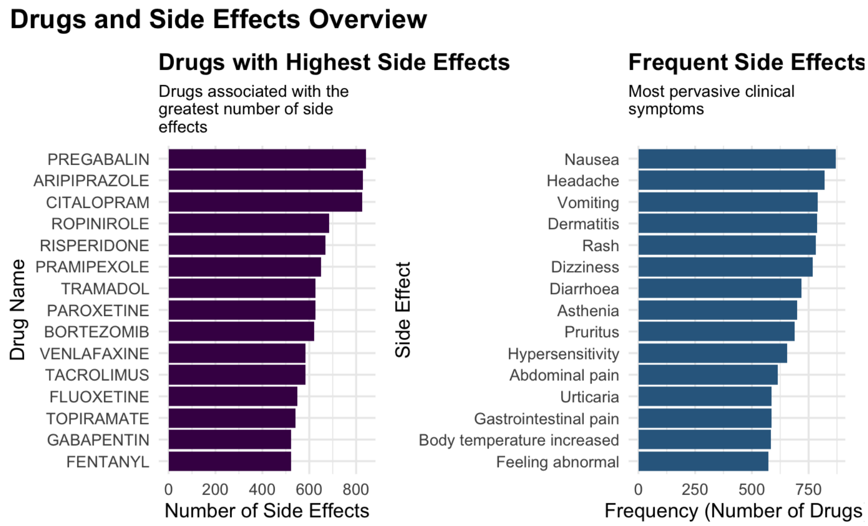

Figure 1: Drugs with most number of side effects and most frequent side effects

The below Figure 2 describes the number of drugs associated with number of side effects. Several drugs are associated with more than 200 side effects and handful of them were associated with as high as 800 side effects.

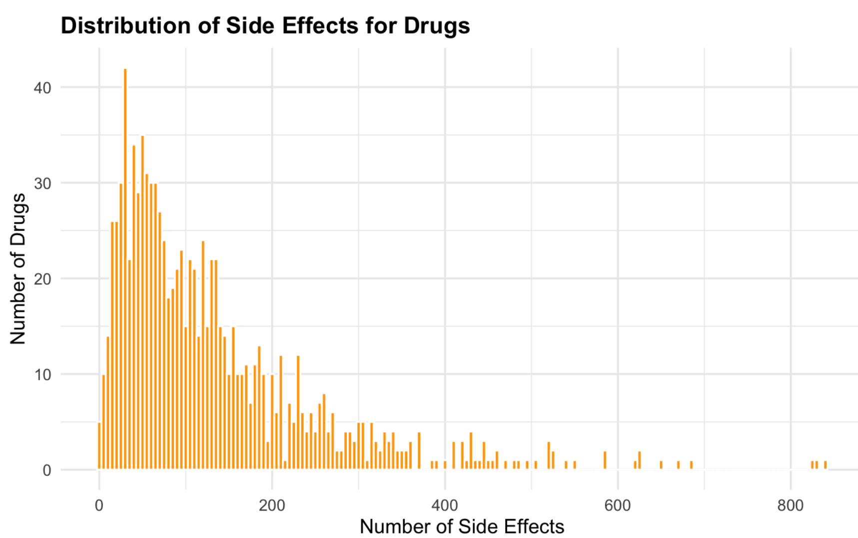

Figure 2: Distribution of side effects

Since most of model functions based on similarity between the drug profiles, the Figure 3 heatmap illustrates the drug similarity based on side effects. Distinct clustering patterns can be observed, suggesting that groups of drugs share similar adverse reaction profiles. The heatmaps also demonstrate the sparsity of the interaction matrix, with positive drug-side effect associations representing only a small fraction of all possible combinations. Overall, the visualisation highlights the heterogeneous nature of ADR profiles and the highly imbalanced characteristics of the prediction task.

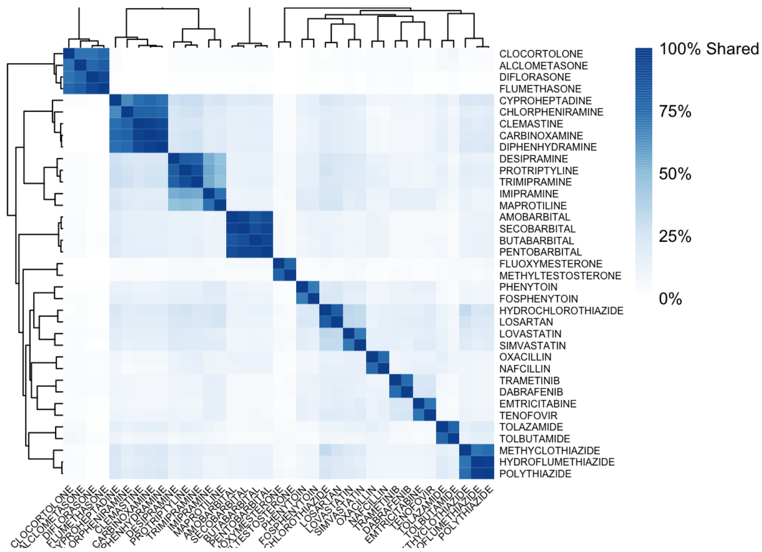

Figure 3: Drug Similarity Heatmap

4.1 Overall Model Performance

Table 1 summarises the predictive performance of all evaluated methods using five-fold cross-validation, where the dataset was randomly partitioned into 75% training data and 25% testing data in each fold. Performance is reported as the mean ± standard deviation of the Area Under the Receiver Operating Characteristic Curve (AUROC) and the Area Under the Precision–Recall Curve (AUPR) across the five folds, with the highest average value for each metric highlighted.

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## Warning: package 'knitr' was built under R version 4.5.2
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Table 2: Mean performance across five-fold cross-validation (AUROC and AUPR)

Method	Mean_AUROC_SD	Mean_AUPR_SD
Naïve Baseline	0.9101 ± 0.0024	0.3584 ± 0.0075
Kernel Regression	0.8745 ± 0.0051	0.4155 ± 0.0051
SVM Linear	0.7955 ± 0.0097	0.3122 ± 0.0124
SVM RBF	0.9001 ± 0.0041	0.3919 ± 0.0068
VKR NMF	0.9183 ± 0.0027	0.4230 ± 0.0033
VKR Weighted NMF	0.8551 ± 0.0035	0.5001 ± 0.0022
VKR SVD	0.9098 ± 0.0033	0.4341 ± 0.0049

Table 2. Summary of predictive performance for all evaluated methods using five-fold cross-validation with a 75% training and 25% testing split, reporting the mean AUROC and AUPR (± standard deviation) across the five folds. The evaluated methods achieved mean AUROC values ranging from 0.7955 ± 0.0097 to 0.9183 ± 0.0027, while mean AUPR values ranged from 0.3122 ± 0.0124 to 0.5001 ± 0.0022.

Among all evaluated approaches, VKR NMF achieved the highest mean AUROC (0.9183 ± 0.0027) together with a mean AUPR of 0.4230 ± 0.0033, indicating the strongest overall discrimination between positive and negative ADR associations.

VKR Weighted NMF produced the highest mean AUPR (0.5001 ± 0.0022), while achieving a mean AUROC of 0.8551 ± 0.0035. VKR SVD also demonstrated strong and consistent performance across all five folds, obtaining a mean AUROC of 0.9098 ± 0.0033 and a mean AUPR of 0.4341 ± 0.0049.

The Naïve Baseline achieved the second-highest mean AUROC (0.9101 ± 0.0024), although its mean AUPR (0.3584 ± 0.0075) was considerably lower than those of the VKR-based methods. Kernel Regression and SVM RBF demonstrated intermediate performance across both evaluation metrics, whereas SVM Linear produced the lowest overall predictive performance, with a mean AUROC of 0.7955 ± 0.0097 and a mean AUPR of 0.3122 ± 0.0124.

4.2 ROC and Threshold-Based Performance Analysis

Figure 4 presents the Receiver Operating Characteristic (ROC) curves for all evaluated methods. VKR NMF achieved the strongest overall discrimination between positive and negative ADR associations, producing the highest AUROC among all evaluated approaches. VKR SVD and the Naïve Baseline also demonstrated competitive ROC performance, whereas SVM Linear exhibited the weakest discrimination. Kernel Regression and SVM RBF showed intermediate performance. Receiver Operating Characteristic (ROC) curves for all evaluated methods. (b) Precision, recall, and F1 score as a function of the decision threshold for the VKR NMF model. The dashed vertical line indicates the optimal threshold (0.2265), corresponding to precision = 0.4355, recall = 0.4760, and F1 = 0.4549.

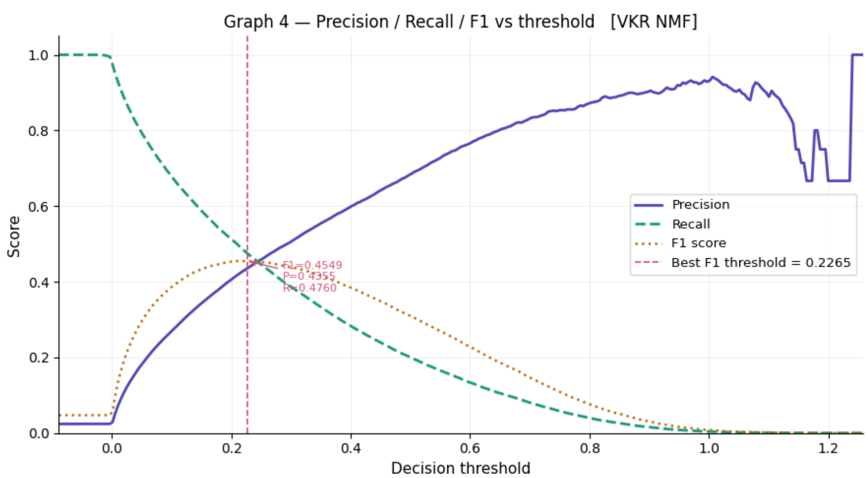

Figure 4: Precision Recall Curve

Figure 5 presents the variation of precision, recall, and F1 score for the VKR NMF model across different decision thresholds. As the decision threshold increases, recall decreases while precision generally increases, illustrating the trade-off between identifying true ADR associations and reducing false positive predictions. The F1 score, which balances precision and recall, reaches its maximum value of 0.4549 at a threshold of 0.2265, corresponding to a precision of 0.4355 and a recall of 0.4760. The optimal threshold is considerably lower than the conventional value of 0.5, indicating that a lower decision boundary provides a more balanced prediction performance for this highly imbalanced ADR dataset.

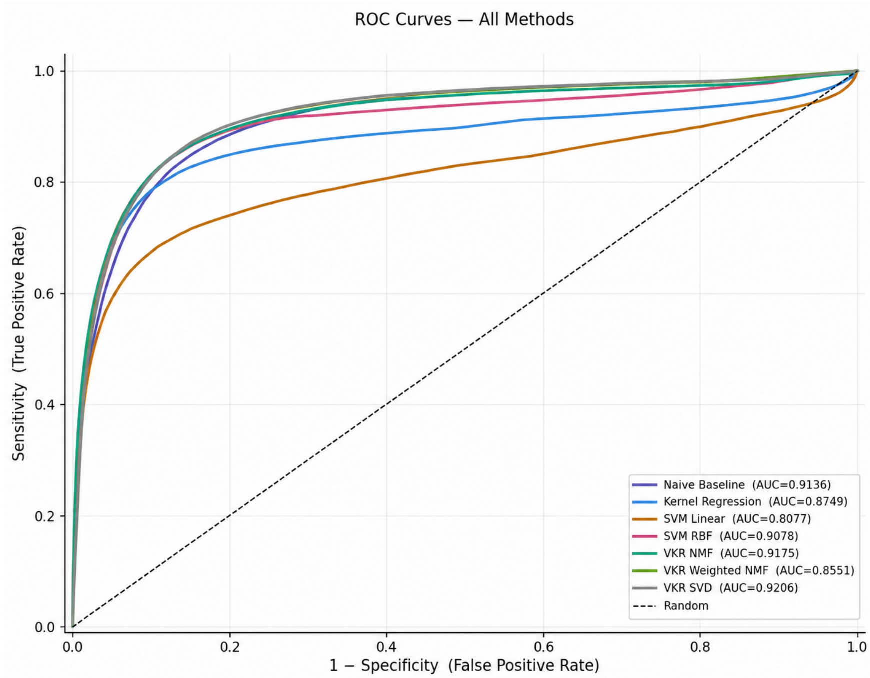

Figure 5: ROC Curve for all methods

4.3 Sensitivity and Specificity Across Decision Thresholds

Figure 6 presents the sensitivity and specificity of each evaluated method across decision thresholds ranging from 0.1 to 0.9. The heatmaps illustrate each model's ability to correctly identify true ADR associations (sensitivity) while correctly rejecting negative associations (specificity). Most models follow the expected pattern in which sensitivity decreases as the decision threshold increases, while specificity increases. The Naïve Baseline, Kernel Regression, SVM variants, VKR NMF, and VKR SVD all exhibit high sensitivity at lower thresholds, followed by a gradual decline as more stringent decision boundaries are applied. By a threshold of 0.5, sensitivity for these models has decreased substantially, whereas specificity exceeds 0.99, indicating a strong ability to avoid false positive predictions. In contrast, VKR Weighted NMF displays a markedly different behaviour. Across the entire threshold range, it maintains substantially higher sensitivity than the remaining methods, achieving 0.976 at a threshold of 0.1 and remaining at 0.918 even at a threshold of 0.9. This high sensitivity is accompanied by comparatively lower specificity, which increases from 0.287 at threshold 0.1 to 0.763 at threshold 0.9. The heatmaps therefore demonstrate the distinct operating characteristics of VKR Weighted NMF relative to the other evaluated approaches.

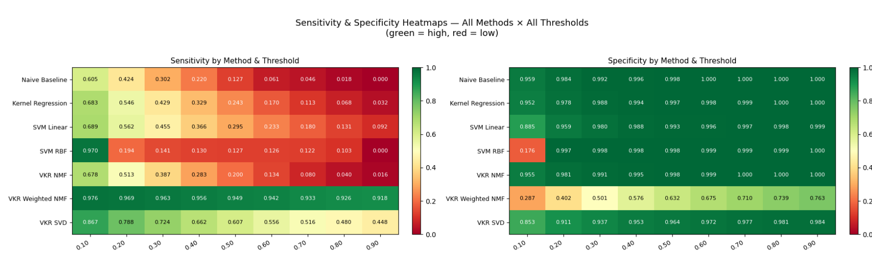

Figure 6: Sensitivity and specificity heatmaps across all methods and decision thresholds (0.1–0.9). Green indicates higher values, whereas red indicates lower values.

Sensitivity and specificity heatmaps across all methods and decision thresholds (0.1–0.9). Green indicates higher values, whereas red indicates lower values.

4.4 Five-Fold Cross-Validation Performance Comparison

Table 3: Five-fold cross-validation results for all methods (AUROC and AUPR)

Method	AUROC_F1	AUROC_F2	AUROC_F3	AUROC_F4	AUROC_F5	AUROC_Mean_SD	AUPR_F1	AUPR_F2	AUPR_F3	AUPR_F4	AUPR_F5	AUPR_Mean_SD
Naive Baseline	0.9089	0.9078	0.9083	0.9116	0.9142	0.9101 ± 0.0024	0.3615	0.3674	0.3485	0.3641	0.3506	0.3584 ± 0.0075
Kernel Regression	0.8676	0.8727	0.8718	0.8787	0.8819	0.8745 ± 0.0051	0.4151	0.4228	0.4090	0.4110	0.4194	0.4155 ± 0.0051
SVM Linear	0.7828	0.8003	0.8000	0.7857	0.8087	0.7955 ± 0.0097	0.3156	0.3243	0.3225	0.2902	0.3085	0.3122 ± 0.0124
SVM RBF	0.8969	0.8974	0.9018	0.8970	0.9074	0.9001 ± 0.0041	0.3945	0.4030	0.3860	0.3923	0.3836	0.3919 ± 0.0068
VKR NMF	0.9158	0.9163	0.9165	0.9204	0.9226	0.9183 ± 0.0027	0.4241	0.4282	0.4182	0.4212	0.4233	0.4230 ± 0.0033
VKR Weighted NMF	0.8510	0.8533	0.8600	0.8587	0.8527	0.8551 ± 0.0035	0.5004	0.5006	0.4960	0.5025	0.5010	0.5001 ± 0.0022
VKR SVD	0.9051	0.9072	0.9106	0.9118	0.9143	0.9098 ± 0.0033	0.4338	0.4427	0.4292	0.4296	0.4353	0.4341 ± 0.0049

Table 3. presents the complete five-fold cross-validation results for all evaluated methods, reporting AUROC and AUPR scores for each fold along with their mean and standard deviation. VKR NMF achieved a mean AUROC of 0.9183 ± 0.0027 and a mean AUPR of 0.4230 ± 0.0033 across the five folds. VKR Weighted NMF obtained a mean AUROC of 0.8551 ± 0.0035 and a mean AUPR of 0.5001 ± 0.0022. VKR SVD recorded a mean AUROC of 0.9098 ± 0.0033 and a mean AUPR of 0.4341 ± 0.0049. The Naïve Baseline achieved a mean AUROC of 0.9101 ± 0.0024 and a mean AUPR of 0.3584 ± 0.0075. Kernel Regression obtained a mean AUROC of 0.8745 ± 0.0051 and a mean AUPR of 0.4155 ± 0.0051. SVM Linear recorded a mean AUROC of 0.7955 ± 0.0097 and a mean AUPR of 0.3122 ± 0.0124, while SVM RBF achieved a mean AUROC of 0.9001 ± 0.0041 and a mean AUPR of 0.3919 ± 0.0068.

4.5 An Interactive Application for ADR Prediction

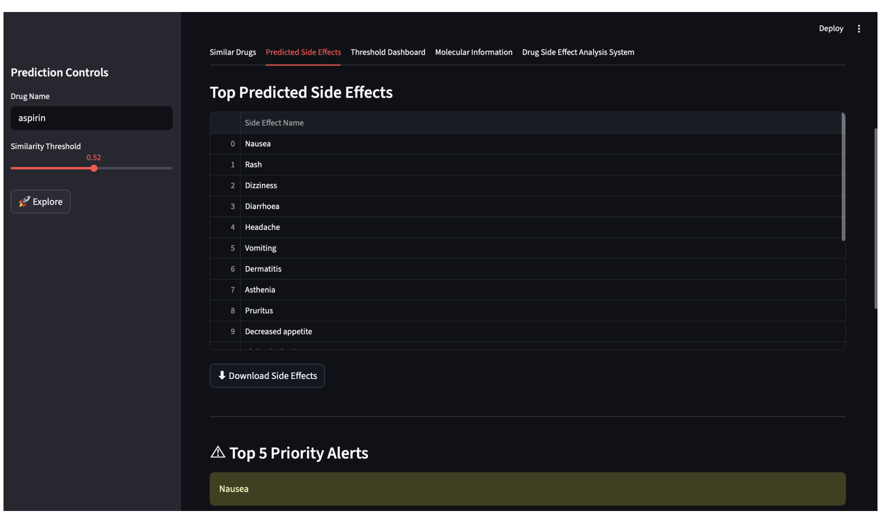

Figure 7: Interactive Application For ADR Prediction

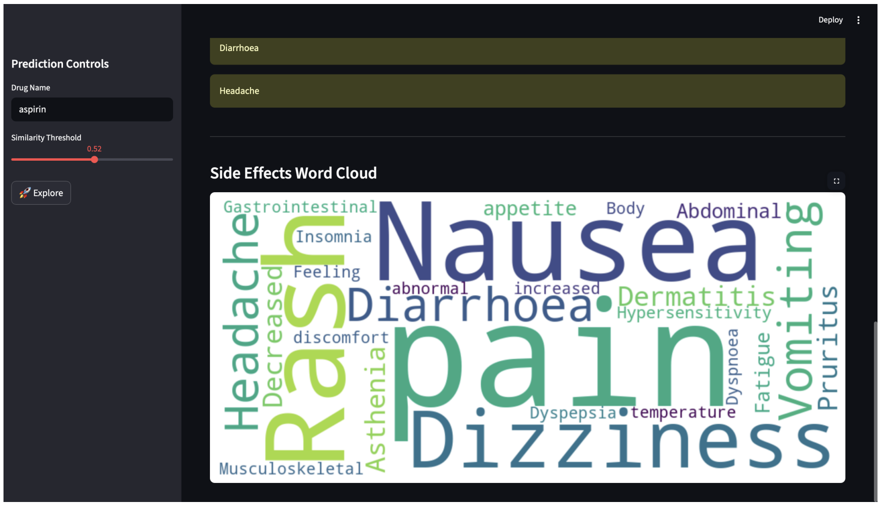

Figure 8: Interactive Application For ADR Prediction

Figure 7 and 8, displays a snapshot of an interactive web application allowing users to input a drug and receive a ranked list of predicted adverse drug reactions from the best-performing model. The app can be run locally where the code, datasets, and deployment steps are all available in the paper's GitHub repository.

5 Discussion

Adverse drug reactions cause significant patient harm and are a major driver of drug withdrawals from the market, and therefore remain a critical concern in pharmacovigilance and drug development ((**Lazarou?** et al. 1998)). Being able to predict them computationally before a drug reaches widespread clinical use is therefore a problem that genuinely matters. This study set out to evaluate whether a range of statistical and machine learning methods, drawing on drug-gene interaction profiles, chemical fingerprints, and side-effect similarity, could predict adverse drug reaction associations in a robust and interpretable way. The results demonstrate that this is feasible, while also revealing important and often underappreciated challenges in how ADR prediction models should be evaluated.

A key point from this work is that the choice of evaluation metric is very important for highly imbalanced biomedical datasets. In this study, AUPR is used as the main metric, while AUROC is treated as a secondary measure. This is because the dataset is heavily imbalanced, with far more negative drug–reaction pairs than positive ones. The Naïve Baseline highlights this issue clearly. Even without using any meaningful drug-specific information, it still achieves an AUROC of over 0.91, which can make it look like a strong model. However, its AUPR is only 0.36, showing that it performs poorly in correctly identifying the rare positive cases. This difference shows that AUROC can sometimes give an overly optimistic view of performance in imbalanced datasets, while AUPR gives a more realistic picture of how well the model actually works in practice. This finding is also in line with previous studies, which suggest that precision–recall based evaluation is more reliable than ROC analysis when the data is highly imbalanced ((**Saito?** and Rehmsmeier 2015)).

The difference between VKR NMF and VKR Weighted NMF shows an important trade-off in ADR prediction. VKR NMF performs best in terms of cross-validated AUROC and gives more precise predictions at standard thresholds. This makes it useful when we need high confidence in the predicted drug–ADR associations. On the other hand, VKR Weighted NMF focuses more on sensitivity across all thresholds. It is able to detect most true ADR associations, even at higher thresholds, but this reduces its specificity, meaning it produces more false positives. In practical use, both models serve different purposes. VKR Weighted NMF is better for early-stage drug safety screening, where it is important not to miss potential risks. VKR NMF is more suitable for later-stage or confirmatory analysis, where higher precision is more important ((**Paul?** et al. 2010)).

The threshold analysis further reinforces an important methodological point: the conventional default threshold of 0.5 is not appropriate for imbalanced ADR datasets. For VKR NMF, the optimal F1 score is achieved at a threshold of 0.2265, which is substantially lower than the default. At this operating point, the model achieves a more balanced trade-off between precision and recall. This indicates that models trained on imbalanced biomedical data tend to produce conservative probability estimates for positive cases, and that threshold calibration is therefore a critical step rather than a post hoc adjustment. Using an uncalibrated threshold such as 0.5 would result in a substantial loss of true positive predictions without meaningful gains in specificity ((**Malone?** et al. 2018)).

From a modelling perspective, the multi-source feature integration approach used in this study seems to work well. Drug–gene interactions, chemical fingerprints, and side-effect mappings each capture different and complementary aspects of drug behaviour. The improvement in AUPR from the Naïve Baseline to the VKR-based models suggests that these features contain real biological signal rather than just statistical patterns. In particular, latent factorisation methods like VKR NMF are able to effectively use these heterogeneous inputs, showing that the learned representation space is informative for predicting adverse drug reactions. An important contribution of the present study is that it builds upon and extends the framework proposed by (**Zhong?** et al.), who demonstrated the effectiveness of combining non-negative matrix factorisation with kernel regression for adverse drug reaction prediction using heterogeneous biological information. While their work established the value of latent factorisation techniques, the current study broadens this approach by evaluating an extended set of biological features, including drug–gene interaction profiles, chemical fingerprints, and side-effect similarity, within a comprehensive comparative framework. In addition, rather than focusing primarily on overall discrimination performance, this study places particular emphasis on the challenges posed by class imbalance by adopting AUPR as the principal evaluation metric and incorporating threshold optimisation into the modelling process. The results obtained are broadly consistent with those of (**Zhong?** et al.), confirming that latent factorisation remains an effective strategy for ADR prediction, while also demonstrating that careful evaluation under imbalanced conditions provides a more realistic assessment of model performance. Together, these methodological refinements and the accompanying interactive implementation represent a practical extension of previous work and contribute to the development of more interpretable and clinically relevant computational pharmacovigilance tools.

Despite these positive results, several limitations should be acknowledged. First, the current framework focuses exclusively on single-drug adverse reactions and does not account for drug-drug interactions, which are known to contribute significantly to real-world adverse events ((**Dechanont?** et al. 2014)). Second, performance may degrade for novel drugs with limited structural similarity or sparse interaction data, since similarity-based approaches rely on sufficient neighbourhood information for generalisation. Third, the training data is derived from pharmacovigilance databases that are inherently biased toward well-studied drugs and frequently reported reactions, introducing reporting bias that may inflate performance on common drug classes while limiting generalisability ((**Hauben?** and Zhou 2012)).

There are several promising directions for future work. Graph neural networks represent a natural extension of the current framework, as they can directly model drug-protein and protein-protein interaction networks in a relational structure rather than relying on flattened feature representations ((**Zitnik?** et al. 2018)). Extending the current framework to drug-drug interaction prediction would also significantly increase its clinical relevance. In addition, integrating continuously updated pharmacovigilance databases would help address temporal drift in drug safety data and improve robustness over time. Finally, incorporating additional replicate datasets and cross-source validation would help reduce dataset-specific bias and improve generalisability across drug classes.

Overall, this study establishes a strong and interpretable baseline framework for ADR prediction with a rigorous evaluation strategy that explicitly accounts for class imbalance. By prioritising AUPR as the main evaluation metric and treating threshold selection as a core modelling decision, the work provides a more realistic assessment of predictive performance than approaches that rely primarily on AUROC. The resulting system is not only analytically robust but also practically usable through the accompanying interactive application, and it provides a solid foundation for future extensions in computational drug safety prediction.

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